

Final Exam List

► The final course grade =

Midterm Exam (30%) + HW (30%) + **Final Exam (15%)**
Final Report (15%) + Attendance (10%)

OS Final Report Topic

- ▶ 1. Topic: I will provide 11 topics for you to choose from.
- ▶ 2. Font Size: 12.
- ▶ 3. Submission Date: Monday of the last week of the semester. Email to me.
- ▶ 4. Number of Pages: 8-10

11 topics list (1)

Chapter 1: Introduction to Operating Systems

1. **Exploring the Interaction Between Hardware-Level Execution Protection (NX/DEP) and Operating System Security Mechanisms**
Exploring how operating systems leverage hardware architecture features to prevent malicious code execution, and the implementation of protections at the system call level in modern OS.
2. **Analysis of Operating System Kernel Design in Heterogeneous Dual-Core Architectures**
Researching how an OS running within a single system (e.g., combining a high-performance processor and a microcontroller) partitions kernel functions and manages hardware resources.
3. **Streamlining Operating Systems in Edge Computing Environments: From Traditional OS to Microkernel Architectures**
Analyzing the evolution of OS architectures and performance trade-offs on resource-constrained edge devices.

11 topics list (2)

Chapter 2: Process Management

4. **Parameter Tuning and Performance Bottleneck Analysis of the Multilevel Feedback Queue (MLFQ) Scheduling Algorithm**
Delving into the design of time quantum and priority demotion mechanisms in MLFQ, and analyzing its scheduling performance under various CPU and I/O workloads.
5. **Comparison of Context Switch Costs Between Lightweight Threads and Traditional Processes Under Modern Computing Workloads**
Analyzing the resource consumption differences when the operating system performs scheduling and state saving for modern multi-threaded applications.
6. **Scheduling Mechanisms in Real-Time Systems: Exploring the Priority Inversion Problem and Its Solutions**
Researching scenarios where high-priority processes are blocked by low-priority ones, and how the OS resolves this through mechanisms like Priority Inheritance.

11 topics list (3)

Chapter 3: Inter-Process Communication and Synchronization

7. **Implementation Analysis of Inter-Process Communication (IPC) in Heterogeneous Computing Environments: A Case Study of the Bridge API Between Linux and Microcontrollers**
Exploring how different processing units or OS environments achieve efficient cross-core data synchronization and signal transmission via specific APIs or shared memory mechanisms.
8. **Application of Operating System Deadlock Prevention and Avoidance Algorithms in Complex System Resource Allocation**
Analyzing the implementation challenges and system overhead of traditional deadlock handling mechanisms, such as the Banker's Algorithm, in modern multitasking environments.
9. **Synchronization Mechanisms for Sensor Data Streams: The Application of Mutexes and Semaphores in Real-Time Stream Processing**
Researching how the OS ensures data consistency in shared buffers when receiving high-frequency external data (e.g., image recognition inputs).

11 topics list (4)

Chapter 4: Memory Management

10. **Architectural Evolution and Hit Rate Analysis of the Translation Lookaside Buffer (TLB) in Modern Hardware Architectures**
Exploring the impact of the OS paging mechanism on memory access speeds, and the critical role of the TLB in reducing the overhead of memory address translation.
11. **The Role and Performance Overhead Evaluation of Extended Page Tables (EPT) in Virtualized Memory Management**
Analyzing the two-stage address translation mechanism from the Guest OS to the host's physical memory in a virtual machine environment, alongside the performance gains brought by hardware-assisted architectures.